

Physics-Informed Intent-Aware Human-Robot Collaborative Transportation With Forward-Looking Adaptation

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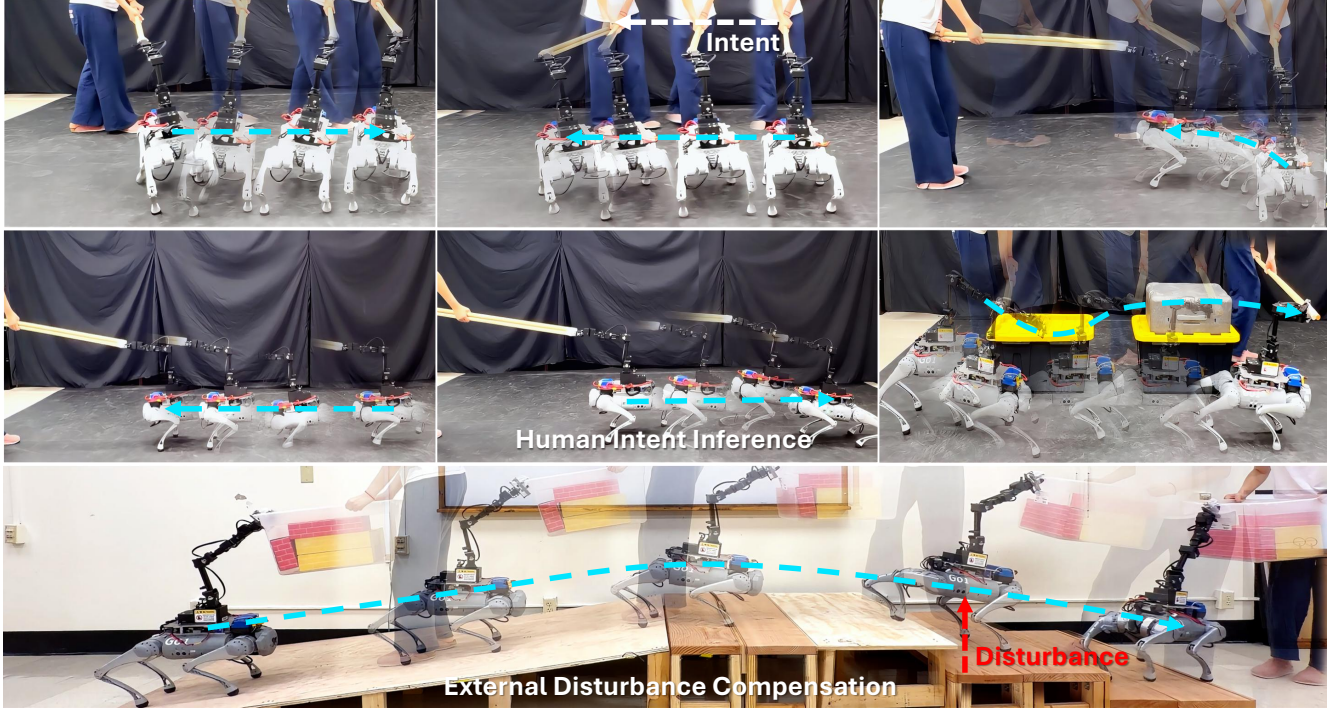


Fig. 1. During human-robot collaborative transportation, the human acts as the leader and initiates the desired motion of the shared payload. The robot should first infer human motion intent from proprioceptive observations to follow the intended motion while compensating for external disturbances to maintain stable transportation.

Abstract—Accurate human-intent following is a fundamental capability for quadruped mobile manipulators in collaborative payload transportation. Existing methods typically infer human intent using direct force sensing or accurate dynamic models, and compensate for disturbances reactively after their effects appear in the robot state. However, two key challenges remain underexplored. First, human intent must be distinguished from non-intent forces, such as external disturbances, using only proprioceptive observations while maintaining consistency with whole-body dynamics. Second, disturbances should be compensated for before they cause deviations from the intended collaborative motion. To address these, we propose Physics-Informed and Forward-looking Adaptation (PIFA), an approach for collaborative transportation. PIFA encodes proprioceptive histories to capture temporal motion and interaction context and jointly predicts human motion intent, interaction wrench, non-intent forces, and the future robot state. These predictions are constrained by the robot’s whole-body dynamics to promote physical consistency. PIFA further converts the predicted intent into coordinated leg and arm commands while using the predicted non-intent forces and future state to compensate for disturbances. Extensive experiments in simulation and on physical robots demonstrate that PIFA enables more accurate human-intent tracking and force estimation while substantially reducing end-effector interaction forces under payload variations and external disturbances compared with existing methods.

I. INTRODUCTION

Quadruped robots have made substantial progress in locomotion and whole-body control, enabling robust mobility across uneven terrain and cluttered environments [1], [2]. Equipping these platforms with manipulator arms further extends their workspace and physical interaction capabilities, enabling a wider range of loco-manipulation tasks [3], [4]. Among these tasks, human–robot collaborative transportation combines the mobility and physical capabilities of quadruped robots with human adaptability and decision-making. In collaborative payload transportation [5], [6], the human typically acts as the leader by initiating the desired motion of the shared payload, while the robot physically collaborates with the human to transport it.

Effective collaborative transportation therefore requires the robot to infer human motion intent [7], [8] while maintaining stable locomotion and manipulation under disturbances [9], [10]. These disturbances include payload weight and load redistribution, payload oscillations, external impacts, and terrain-induced body motion [11], [12]. As illustrated in Figure 1, these two requirements are closely coupled but

demand fundamentally different responses: interaction forces arising from human intent should be followed to achieve the desired payload motion, whereas external disturbances should be compensated for to preserve stability. Failing to distinguish them may cause the robot to resist the human's intended motion or treat disturbances as intentional commands, degrading both coordination and stability. However, human intent and external disturbances can induce similar changes in the robot's proprioceptive observations, making this distinction challenging.

Given the importance, classical methods estimate human interaction forces either directly using force/torque sensors [13], [14] or indirectly from robot model measurements [15], [16], and use them to follow human motion through impedance [17] or admittance control [18]. However, they either require additional sensing or rely on accurate robot models, whose estimation accuracy degrades under payload variations and external disturbances. Learning-based approaches [19], [20] reduce reliance on direct force sensing by inferring human intent from proprioceptive and temporal observations [21], [22]. However, human intent and external disturbances can produce similar proprioceptive responses, making them difficult to distinguish without a dynamics model and limiting the robot's ability to follow intended human motion while compensating for disturbances. Quadruped locomotion robustness has been improved through adaptive whole-body control [23], latent dynamics estimation [24], and learned disturbance adaptation [25]. However, they typically compensate for disturbances only after their effects appear in the robot state, leading to delayed correction and larger deviations from the intended collaborative motion. Overall, existing approaches generally do not jointly distinguish human intent from non-intent interaction effects without direct force sensing, enable forward-looking disturbance compensation, and enforce consistency with whole-body dynamics during collaborative transportation.

To address these challenges, we introduce Physics-Informed and Forward-looking Adaptation (PIFA), an approach for human-robot collaborative transportation. PIFA first encodes a history of proprioceptive states into a latent representation that captures the robot's motion and interaction context. A forward dynamics predictor then jointly predicts human motion intent, interaction wrench, non-intent forces, and the future robot state. These predictions are constrained by the robot's whole-body dynamics to promote physical consistency (*physics-informed*). The predicted human intent and interaction wrench are mapped to coordinated base-following commands and compliant end-effector motion, while the predicted non-intent forces and future robot state are used for whole-body compensation. In parallel, the predicted future robot state is compared with the intended motion to identify disturbance-induced deviations before they appear in the measured state (*forward-looking*). These predicted deviations are then used to generate whole-body compensation, allowing the controller to preserve the desired human-following behavior under changing payload dynamics and external disturbances.

The main contributions of this work are twofold:

- We introduce a novel physics-informed method that jointly predicts human motion intent, interaction wrench, non-intent forces, and the future robot state from proprioceptive histories, while constraining these predictions through whole-body dynamics. This enables data-driven intent and force estimation without direct force sensing while ensuring physical consistency under time-varying human intent and external disturbances.
- We propose a novel forward-looking locomotion controller for collaborative transportation, which converts predicted human intent and non-intent forces into coordinated leg and arm commands before the resulting motion deviations occur, thus enabling more accurate intent following, timely disturbance compensation, and substantially lower end-effector interaction forces under payload variations and external disturbances.

II. RELATED WORK

A. Human-Robot Collaborative Transportation

Classical physical human-robot interaction methods commonly rely on impedance control [17], [26] or admittance control [3], [8] to establish compliant interaction between the human and robot, enabling leader-follower transportation. Building on these methods, classical approaches have explored intent-aware collaborative transportation by interpreting physical interaction as an indication of human intent. Such approaches typically obtain interaction information either through direct force/torque sensing or model-based estimation. Direct sensing methods use force/torque sensors or integrated joint-torque sensors to measure physical interaction between the human and robot, but require additional sensing hardware and calibration [13], [14]. Alternatively, model-based approaches [15], [16] estimate interaction forces from robot dynamics and proprioceptive measurements using momentum observers [27], disturbance observers [28], or joint-torque-based estimators [29]. However, these methods can suffer from degraded estimation accuracy under model uncertainty, payload variations, and external disturbances.

Learning-based methods have been introduced to infer human intent and support diverse human-robot collaboration tasks. Several approaches estimate human intent from measured interaction forces [30], human motion [31]. While these methods can capture nonlinear relationships between human intent and collaborative motion, they still rely on dedicated external sensing modalities, such as force/torque sensors, cameras, or motion-capture systems. More recent approaches infer human intent directly from robot motion or proprioceptive histories, reducing the need for external force/torque sensing [19]–[21]. Temporal observations allow learned models to capture latent interaction patterns that may not be observable from an instantaneous robot state alone. However, human intent and non-intent disturbances can induce similar robot motions and proprioceptive responses. Consequently, proprioceptive observations alone cannot reliably distinguish robot motion changes induced by human intent from those caused by external disturbances.

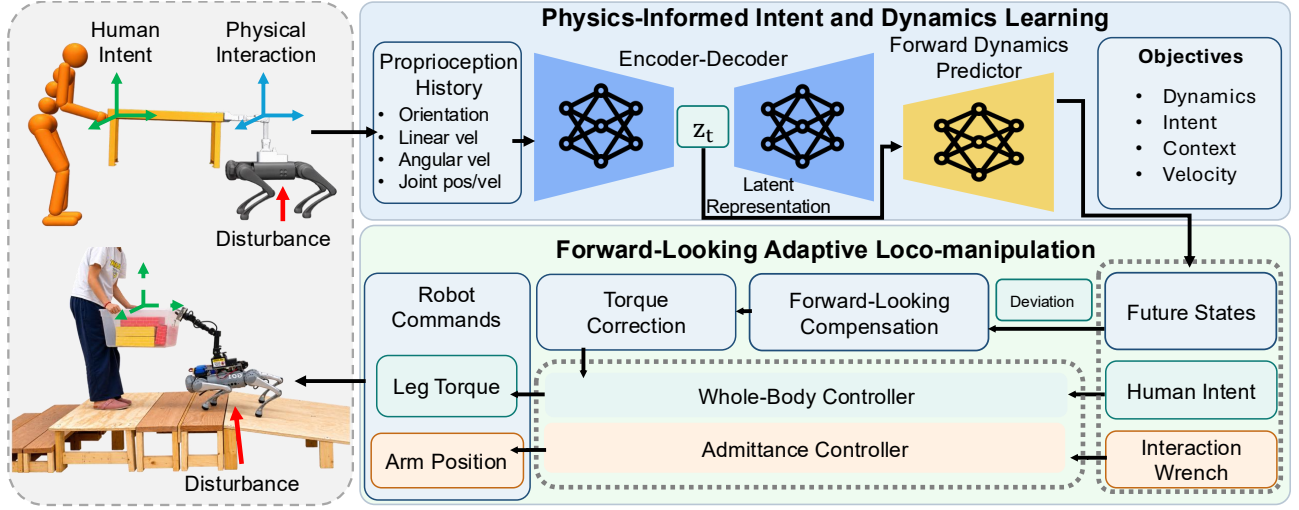


Fig. 2. PIFA encodes proprioceptive histories into a latent representation and uses a physics-informed forward dynamics predictor to estimate future robot states, human intent, interaction wrench, and non-intent forces under whole-body dynamics constraints. The predictions are then used by a forward-looking adaptive loco-manipulation controller to generate whole-body torque compensation and compliant arm commands under external disturbances.

B. Quadruped Locomotion under External Disturbance

Recent advances in quadruped locomotion have enabled robust locomotion and loco-manipulation in challenging environments, including uneven terrain, stairs, and other complex surfaces. Prior work improves robustness through disturbance observers, adaptive whole-body control [24], latent dynamics estimation [32], and learned disturbance adaptation [25]. These methods can effectively compensate for unknown external disturbances. Physics-informed neural networks (PINNs) have also gained increasing attention in robotics [33], with applications in manipulator dynamics [34], quadrotor dynamics [35], and motion planning [36]. PINNs incorporate known physical laws, such as robot equations of motion, into the learning objective to promote physically consistent predictions and reduce reliance on purely data-driven correlations. However, most existing methods are reactive, responding only after disturbances have altered the robot's motion, which can delay corrective action and increase deviations from the intended behavior.

III. APPROACH

A. Problem Definition

We consider a collaborative transportation task in which a human and a quadruped robot equipped with a manipulator jointly carry a payload, with the human leading the task through physical interaction.

We define robot proprioceptive measurements, including the position and orientation of the robot mobile base as $\mathbf{p}$ and $\boldsymbol{\theta}$, the positions of the leg and arm joints as $\mathbf{q}$ and $\boldsymbol{\eta}$, as well as the leg and arm joint torques as $\boldsymbol{\tau}$ and $\boldsymbol{\mu}$ separately. Given the proprioceptive measurements, we construct robot configuration representation as $\boldsymbol{\rho} = [\mathbf{p}, \boldsymbol{\theta}, \mathbf{q}, \boldsymbol{\eta}]^\top$ with $\dot{\boldsymbol{\rho}}$ and $\ddot{\boldsymbol{\rho}}$ denoting the corresponding velocity and acceleration separately, and robot state representation as $\mathbf{s} = [\mathbf{q}, \dot{\mathbf{q}}, \boldsymbol{\eta}, \dot{\boldsymbol{\eta}}, \dot{\boldsymbol{\rho}}, \ddot{\boldsymbol{\rho}}]^\top$.

Given the robot configuration and state representations, we aim to predict 1). the human intent, denoted as $\mathbf{b} =$

$[v_x, v_y, v_z, \omega_z]^\top$ with v_x , v_y , and v_z denoting the human-intent linear velocities and ω_z denoting the yaw angular velocity. 2). the interaction wrench applied to the end effector, denoted as $\mathbf{w} = [w_x, w_y, w_z]^\top$, with w_x , w_y , and w_z denoting the force on x, y, z directions. 3). the control command to robot legs $\boldsymbol{\tau}^{\text{cmd}}$ in force space and the control command to robot arm $\mathbf{a}^{\text{cmd}}$ in position space.

During collaborative transportation, the robot is subject to 1). leg and arm actuation forces $\mathbf{S}_l^\top \boldsymbol{\tau}$ and $\mathbf{S}_a^\top \boldsymbol{\mu}$, where $\mathbf{S}_l$ and $\mathbf{S}_a$ are fixed selection matrices for the leg and arm joints, respectively; 2). ground-contact force $\mathbf{J}_c^\top \mathbf{f}$, where $\mathbf{J}_c$ is the corresponding Jacobian and $\mathbf{f}$ denotes the ground-contact force; 3). human interaction force $\mathbf{J}^\top \mathbf{w}$; and (4) external disturbance $\mathbf{Q}^d$. Therefore, the non-intent force is defined as $\mathbf{Q} \triangleq \mathbf{S}_l^\top \boldsymbol{\tau} + \mathbf{S}_a^\top \boldsymbol{\mu} + \mathbf{J}_c^\top \mathbf{f} + \mathbf{Q}^d$. The resulting whole-body dynamics of the quadruped robot with an arm are given by

$$\mathbf{M}(\boldsymbol{\rho})\ddot{\boldsymbol{\rho}} + \mathbf{h}(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}) = \mathbf{Q} + \mathbf{J}^\top \mathbf{w}, \quad (1)$$

where $\mathbf{M}(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ is the configuration-dependent mass matrix, i.e., $\mathbf{M}(\boldsymbol{\rho})$ is its value evaluated at the current configuration $\boldsymbol{\rho}$. The vector-valued map $\mathbf{h}(\cdot) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ collects the Coriolis, centrifugal, and gravitational bias forces, evaluated at $(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}})$.

Given the robot configuration and state representations, as well as the robot dynamics, we aim to enable effective human-robot collaborative transportation by addressing the following problems. The overall approach is shown in Figure 2.

- The robot should accurately predict the human's intended motion and distinguish it from non-intent effects, such as terrain-induced disturbances, while satisfying physically consistent robot whole-body dynamics.
- The robot should coordinate its mobile base, legs, and manipulator to follow the human's intent while compensating for non-intent forces, thereby maintaining stable whole-body motion and payload transportation.

B. Physics-Informed Intent and Dynamics Learning

Without direct interaction-force sensing, human motion intent and non-intent forces (e.g., external disturbances) can induce similar proprioceptive responses, making it difficult for the robot to follow human intent while compensating for disturbances during whole-body motion. We are inspired by the observation that, although these forces may be indistinguishable from an instantaneous robot state, they often exhibit different temporal effects, where human intent typically induces persistent and directionally coherent motion, whereas disturbances tend to cause abrupt deviations from the recent motion trend. Therefore, we encode a history of proprioceptive states to jointly infer human intent, non-intent forces, and the corresponding robot state transition under whole-body dynamics constraints.

Specifically, we construct a history vector from the most recent L robot states as $\mathcal{H}_t = [\mathbf{s}_{t-L+1}^\top, \dots, \mathbf{s}_t^\top]^\top$, where L denotes the history length. We design a history encoder ϕ that maps $\mathcal{H}_t$ to a latent representation $\mathbf{z}_t$. To learn a compact and informative latent representation of the recent robot dynamics, we pair the encoder ϕ with a decoder ψ to reconstruct the history states from the latent representation, defined as $\hat{\mathcal{H}}_t = \psi(\mathbf{z}_t)$. Both ϕ and ψ are implemented as multilayer perceptrons (MLP) with ELU activations. We define the context-encoding objective for ϕ and ψ as

$$\mathcal{L}_{\text{CE}} = \|\mathcal{H}_t - \hat{\mathcal{H}}_t\|_2^2 + D_{\text{KL}}(q(\mathbf{z}_t|\mathcal{H}_t) \|\mathcal{N}(\mathbf{0}, \mathbf{I})) \quad (2)$$

where the first term encourages $\mathbf{z}_t$ to preserve information from the state history, while the second regularizes the latent representation toward a standard Gaussian distribution, encouraging a bounded and smoothly structured representation.

Given the latent context $\mathbf{z}_t$ and the current robot state $\mathbf{s}_t$, we design a forward dynamics predictor f_θ to jointly predict the future robot state $\hat{\mathbf{s}}_{t+1}$, non-intent force $\hat{\mathbf{Q}}_{t+1}$, human motion intent $\hat{\mathbf{b}}_t$, and interaction wrench $\hat{\mathbf{w}}_t$:

$$(\hat{\mathbf{s}}_{t+1}, \hat{\mathbf{Q}}_{t+1}, \hat{\mathbf{b}}_t, \hat{\mathbf{w}}_t) = f_\theta(\mathbf{s}_t, \mathbf{z}_t) \quad (3)$$

where f_θ is implemented as an MLP with ELU activations.

As the predicted state and force quantities jointly characterize the robot state transition, they should satisfy the whole-body dynamics in Eq. 1 to promote physically consistent predictions. We define a whole-body dynamics objective at timestep $t+1$ by minimizing the corresponding residual:

$$\mathcal{L}_{\text{wb}} = \|\mathbf{M}(\hat{\mathbf{p}}_{t+1})\ddot{\hat{\mathbf{p}}}_{t+1} + \mathbf{h}(\hat{\mathbf{p}}_{t+1}, \dot{\hat{\mathbf{p}}}_{t+1}) - \hat{\mathbf{Q}}_{t+1} - \mathbf{J}_{t+1}^\top \hat{\mathbf{w}}_t\|_2^2 \quad (4)$$

where $\mathbf{M}(\hat{\mathbf{p}}_{t+1})\ddot{\hat{\mathbf{p}}}_{t+1}$ and $\mathbf{h}(\hat{\mathbf{p}}_{t+1}, \dot{\hat{\mathbf{p}}}_{t+1})$ characterize the predicted whole-body motion; $\hat{\mathbf{p}}_{t+1}$, $\dot{\hat{\mathbf{p}}}_{t+1}$, and $\ddot{\hat{\mathbf{p}}}_{t+1}$ denote the predicted robot configuration, velocity, and acceleration, respectively; $\hat{\mathbf{Q}}_{t+1}$ denotes the predicted non-intent force; and $\hat{\mathbf{w}}_t$ denotes the predicted interaction wrench. To explicitly learn human motion intent and distinguish it from non-intent forces, we define the prediction objective as

$$\mathcal{L}_{\text{pred}} = \|\mathbf{s}_{t+1} - \hat{\mathbf{s}}_{t+1}\|_2^2 + \|\mathbf{Q}_{t+1} - \hat{\mathbf{Q}}_{t+1}\|_2^2 + \|\mathbf{b}_t - \hat{\mathbf{b}}_t\|_2^2 + \|\mathbf{w}_t - \hat{\mathbf{w}}_t\|_2^2 \quad (5)$$

Since accurate joint velocities are important for evaluating the whole-body dynamics, we additionally emphasize their prediction accuracy through a velocity objective

$$\mathcal{L}_{\text{vel}} = \|\dot{\mathbf{q}}_{t+1} - \dot{\hat{\mathbf{q}}}_{t+1}\|_2^2 + \|\dot{\mathbf{n}}_{t+1} - \dot{\hat{\mathbf{n}}}_{t+1}\|_2^2 \quad (6)$$

Finally, the overall physics-informed learning objective is defined as $\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{wb}} + \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{vel}}$. Intuitively, the history encoder ϕ captures the robot's evolving motion under human intent, which is guided by $\mathcal{L}_{\text{CE}}$. The forward dynamics predictor f_θ can therefore attribute temporally persistent changes to human intent, while assigning abrupt deviations from this trend to non-intent forces. The objectives $\mathcal{L}_{\text{pred}}$ and $\mathcal{L}_{\text{vel}}$ guide the separation between intent and non-intent forces, and the whole-body dynamics objective $\mathcal{L}_{\text{wb}}$ ensures that the predicted intent-related wrench, non-intent forces, and robot motion jointly provide a physically consistent explanation of the observed state transition. Our unified PIFA approach is trained end-to-end by backpropagating the gradients of the physics-informed loss through the entire model, jointly updating the history encoder ϕ , decoder ψ , and forward dynamics predictor f_θ .

C. Forward-Looking Adaptive Loco-manipulation

Human-robot collaborative transportation requires the robot to follow the human's intended motion compliantly. Meanwhile, external disturbances can drive the robot away from this desired motion. Therefore, we must compensate for it before the deviation occurs.

To address this challenge, PIFA uses the forward dynamics predictor f_θ to infer the human motion intent $\hat{\mathbf{b}}_t$ and interaction wrench $\hat{\mathbf{w}}_t$. The inferred intent serves as the desired base-velocity command for compliant human following, while the interaction wrench is converted through the admittance model into compliant arm motion. Meanwhile, PIFA uses the predicted future robot state $\hat{\mathbf{s}}_{t+1}$ to estimate the next-step joint deviation, which is used by the adaptive law to generate proactive leg-torque compensation.

Specifically, we use the predicted human motion intent $\hat{\mathbf{b}}_t$ as the desired base-velocity command, enabling compliant base following of the human motion. Given $\hat{\mathbf{b}}_t$ and the current robot state $\mathbf{s}_t$, a nominal whole-body controller π is used, such as Whole-Body Impulse Control (WBIC) [37], to generate the desired leg position and torque commands as

$$(\mathbf{q}_{t+1}^d, \boldsymbol{\tau}_t^d) = \pi(\mathbf{s}_t, \hat{\mathbf{b}}_t) \quad (7)$$

where $\mathbf{q}_{t+1}^d$ and $\boldsymbol{\tau}_t^d$ denote the desired joint-position and torque commands, respectively.

Disturbances can drive the robot motion away from the desired behavior. We therefore define the predicted next-step joint-position deviation as $\Delta \mathbf{q}_{t+1} = \mathbf{q}_{t+1}^d - \hat{\mathbf{q}}_{t+1}$, where $\hat{\mathbf{q}}_{t+1}$ is the predicted leg joint position extracted from the future state $\hat{\mathbf{s}}_{t+1}$. We use this deviation to determine an adaptive compensation gain through a bounded-gain-forgetting (BGF) composite adaptive law [38]–[40]. Formally, the rate of change of the compensation gain $\mathbf{G}_t$ is given by

$$\dot{\mathbf{G}}_t = -\mathbf{R}_t \Delta \mathbf{q}_{t+1} (\mathbf{e}_q^\top + \kappa \mathbf{e}_\tau^\top) \quad (8)$$

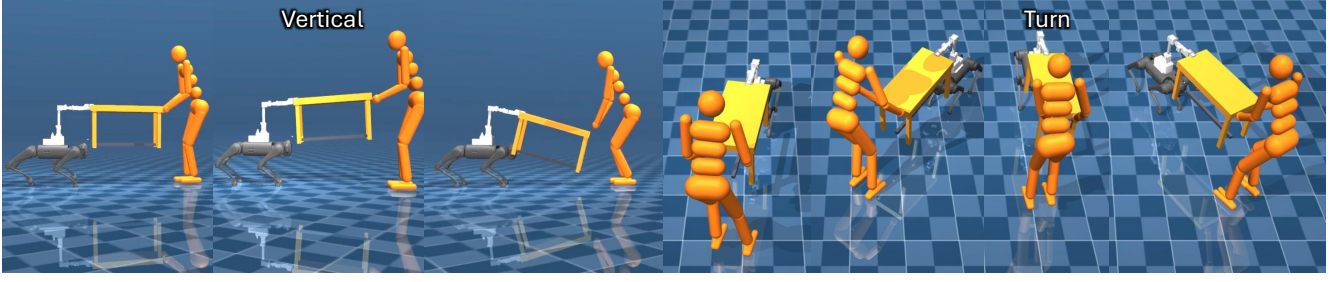


Fig. 3. Qualitative results in MuJoCo simulation. A Unitree Go1 quadruped equipped with a K1 arm collaboratively transports a shared payload with a virtual human partner. The virtual partner generates diverse motion intents, including vertical, horizontal, and rotational motions under external disturbances.

where $e_q = \dot{\mathbf{q}}_t - \alpha (\mathbf{q}_t^d - \mathbf{q}_t)$ and $e_\tau = \tau_t^d - \tau_t$ denote the joint-state and torque tracking errors, respectively, while $\alpha > 0$ and $\kappa > 0$ are hyperparameters that weight the joint-position tracking contribution and torque tracking error, respectively. The adaptation rate is governed by the matrix $\mathbf{R}_t$, which is updated according to

$$\dot{\mathbf{R}}_t = \lambda_t \mathbf{R}_t - \mathbf{R}_t \Delta \mathbf{q}_{t+1} \Delta \mathbf{q}_{t+1}^\top \mathbf{R}_t, \quad (9)$$

where λ_t is the forgetting factor. At each control step, the adaptive gains are updated over the elapsed time step Δt as

$$\mathbf{G}_{t+1} = \mathbf{G}_t + \Delta t \dot{\mathbf{G}}_t, \quad \mathbf{R}_{t+1} = \mathbf{R}_t + \Delta t \dot{\mathbf{R}}_t. \quad (10)$$

Finally, the resulting adaptive gain is applied to the predicted joint deviation to generate the proactive torque compensation $\Delta \tau_t = \mathbf{G}_t \Delta \mathbf{q}_{t+1}$. This correction is then added to the desired torque command, yielding the compensated leg-joint torque command

$$\tau_t^{\text{cmd}} = \tau_t^d + \Delta \tau_t. \quad (11)$$

For compliant arm manipulation, we adopt a second-order admittance model [41] to convert the predicted interaction wrench $\hat{\mathbf{w}}_t$ into a compliant end-effector offset $\Delta \mathbf{x}_t$:

$$\mathbf{M}_a \Delta \ddot{\mathbf{x}}_t + \mathbf{D}_a \Delta \dot{\mathbf{x}}_t + \mathbf{K}_a \Delta \mathbf{x}_t = \hat{\mathbf{w}}_t, \quad (12)$$

where $\mathbf{M}_a$, $\mathbf{D}_a$, and $\mathbf{K}_a$ denote the virtual inertia, damping, and stiffness matrices, respectively. During deployment, we provide a desired grasp pose $\mathbf{g}^d$, and compute the vertical end-effector offset as $\Delta \mathbf{x}_{z,t} = \mathbf{c} \hat{v}_{z,t} \Delta t$, where $\hat{v}_{z,t}$ is the predicted vertical intent velocity obtained from $\hat{\mathbf{b}}_t$, and $\mathbf{c}$ denotes the unit vector along the vertical direction. Finally, the desired end-effector pose is then constructed as

$$\mathbf{x}_t^d = \mathbf{g}^d + \Delta \mathbf{x}_t + \Delta \mathbf{x}_{z,t}. \quad (13)$$

Finally, $\mathbf{x}_t^d$ is converted into the arm joint-position command through inverse kinematics as

$$\mathbf{a}_t^{\text{cmd}} = f_{\text{IK}}(\boldsymbol{\eta}_t, \mathbf{x}_t^d), \quad (14)$$

where $\mathbf{a}_t^{\text{cmd}}$ is tracked by the low-level arm controller.

IV. EXPERIMENTS

A. Experimental Setups

We comprehensively evaluate PIFA in both simulation and real-world settings, including (1) MuJoCo simulation [43] and (2) a physical Unitree GO1 quadruped equipped with a K1 arm. In all scenarios, the human acts as the leader and initiates the desired motion of the shared payload, while the robot infers the human motion intent to follow the intended motion and compensates for external disturbances to maintain stable transportation. The evaluation spans varying payloads, human intent patterns, terrain conditions, and external disturbances to assess human-intent estimation, human-motion tracking, disturbance compensation, and overall system robustness.

Training data are collected in simulation using a Unitree Go1 equipped with a K1 arm while collaboratively transporting a rigid payload with a virtual partner acting at the opposite end. The virtual partner is commanded by $\mathbf{b}_t$, while the interaction wrench $\mathbf{w}_t$ arises naturally from the welded grasp constraint. The command $\mathbf{b}_t$ is used only to drive the virtual partner and supervise the intent prediction; it is neither provided to the robot controller nor included in the robot observation. During data collection, the quadruped base is controlled by a classical admittance controller, while the manipulator executes compliant position commands.

Each episode samples a partner motion from a predefined set of motion primitives, with the payload mass uniformly sampled from $[0.6, 1.2]$ kg. External disturbances are introduced independently through randomized base pushes and perturbations. Each episode lasts 30 s, with each training sample containing a 1.0 s history of proprioceptive observations. During testing, PIFA generates locomotion commands at 0.5 kHz, while the nominal locomotion controller and BGF update law compute compensation terms at 1 kHz. Each testing trial consists of 15,000 control steps, corresponding to 15 s of continuous operation at a 1 kHz sampling rate. Further implementation and experimental details are provided in the supplementary material.

We compare PIFA against two prior methods and two baselines: (1) *Damped Arm (DA)* [42], which uses classical PD control with the manipulator operated in damping mode, where human interaction induces end-effector motion that is mapped to base-motion commands through the controller; (2) *Reinforcement Learning (TD3+BC)* [22], which learns a

TABLE I

QUANTITATIVE COMPARISON UNDER DIFFERENT PAYLOAD WEIGHTS AND EXTERNAL DISTURBANCES. LOWER VALUES ARE BETTER FOR ALL METRICS.

Method	Payload Variation							
	1.0 kg				1.5 kg			
	Lin. MAE ↓ (m/s)	Ang. MAE ↓ (rad/s)	F-est. MAE ↓ (N)	EE Force ↓ (N)	Lin. MAE ↓ (m/s)	Ang. MAE ↓ (rad/s)	F-est. MAE ↓ (N)	EE Force ↓ (N)
DA [42]	0.2678 ± 0.0035	0.1068 ± 0.0011	4.461 ± 0.060	6.62 ± 0.07	0.2714 ± 0.0006	0.1082 ± 0.0005	4.467 ± 0.002	6.66 ± 0.02
TD3+BC [22]	0.0619 ± 0.0006	0.0684 ± 0.0008	1.802 ± 0.008	4.18 ± 0.02	0.0617 ± 0.0004	0.0682 ± 0.0006	1.817 ± 0.007	4.21 ± 0.02
PIFA-NW	0.0537 ± 0.0009	0.0528 ± 0.0004	1.532 ± 0.014	3.10 ± 0.03	0.0546 ± 0.0015	0.0531 ± 0.0005	1.526 ± 0.015	3.09 ± 0.01
PIFA-NP	0.0499 ± 0.0005	0.0517 ± 0.0015	1.402 ± 0.009	2.97 ± 0.01	0.0499 ± 0.0002	0.0509 ± 0.0005	1.402 ± 0.015	2.96 ± 0.02
PIFA (ours)	0.0508 ± 0.0025	0.0423 ± 0.0009	1.412 ± 0.212	2.85 ± 0.12	0.0490 ± 0.0001	0.0416 ± 0.0002	1.326 ± 0.002	2.78 ± 0.01
Method	External Disturbance							
	Torso Disturbance				Torso + Arm Disturbance			
	Lin. MAE ↓ (m/s)	Ang. MAE ↓ (rad/s)	F-est. MAE ↓ (N)	EE Force ↓ (N)	Lin. MAE ↓ (m/s)	Ang. MAE ↓ (rad/s)	F-est. MAE ↓ (N)	EE Force ↓ (N)
DA [42]	0.2661 ± 0.0014	0.1042 ± 0.0007	4.426 ± 0.003	10.64 ± 0.02	0.2695 ± 0.0045	0.1103 ± 0.0048	4.424 ± 0.060	10.47 ± 0.20
TD3+BC [22]	0.0614 ± 0.0003	0.0675 ± 0.0006	1.795 ± 0.003	4.18 ± 0.00	0.0639 ± 0.0021	0.0698 ± 0.0010	1.909 ± 0.066	4.36 ± 0.13
PIFA-NW	0.0541 ± 0.0011	0.0534 ± 0.0007	1.538 ± 0.009	3.08 ± 0.02	0.0540 ± 0.0036	0.0526 ± 0.0022	1.515 ± 0.043	3.24 ± 0.10
PIFA-NP	0.0502 ± 0.0004	0.0513 ± 0.0004	1.416 ± 0.010	2.98 ± 0.01	0.0530 ± 0.0014	0.0527 ± 0.0021	1.470 ± 0.034	3.04 ± 0.02
PIFA (ours)	0.0487 ± 0.0001	0.0415 ± 0.0003	1.324 ± 0.005	2.78 ± 0.02	0.0511 ± 0.0016	0.0435 ± 0.0012	1.410 ± 0.020	2.98 ± 0.07

collaborative control policy from offline demonstrations using TD3 with behavior-cloning regularization; (3) *PIFA without Physics* (PIFA-NP), a baseline that removes the whole-body dynamics loss in Eq. 4, resulting in a purely data-driven model; and (4) *PIFA without Interaction Wrench* (PIFA-NW), a baseline that removes the interaction wrench term $\mathbf{J}^\top \mathbf{w}$ from Eq. 4 to evaluate the contribution of explicitly modeling interaction forces.

We quantitatively evaluate PIFA and the comparison methods using three metrics: (1) *Intent Tracking Error* [22], reported separately as linear (**Lin. MAE**) and angular (**Ang. MAE**) mean absolute errors between the human motion intent and the robot’s executed linear and angular velocities, measured in m/s and rad/s, respectively; (2) *Force Estimation Error* (**F-est. MAE**) [21], measured as the MAE between the predicted and ground-truth interaction forces, in N; and (3) *End-Effector Interaction Force* (**EE Force**) [22], measured as the average magnitude of the horizontal interaction force at the end-effector, in N, reflecting the human effort required to guide the robot during collaboration.

B. Collaborative Transportation in Simulation

We first evaluate PIFA in MuJoCo simulation under different payload and disturbance conditions. In each scenario, the quadruped manipulator collaboratively transports a shared payload with a virtual human partner and adapts its leg and arm motions to follow the partner’s commanded movement. The payload mass is varied to evaluate performance under changes in the coupled robot-payload dynamics, while external forces are applied to assess robustness to disturbances.

The qualitative results are shown in Figure 3. PIFA enables stable collaborative transportation under diverse human motions, including horizontal rotation and vertical motion. During vertical collaboration, the robot adapts its arm to changes in payload height. During turning, it continuously adjusts its locomotion and manipulator configuration to follow changes in payload orientation. Notably, the robot remains

stable under continuous randomized torso and arm disturbances, demonstrating the effectiveness of physics-informed prediction and forward-looking compensation for coordinated whole-body motion during collaborative transportation.

The quantitative results are shown in Table I, with standard deviations computed over three independent runs. Under the 1.0 kg payload, DA and TD3+BC exhibit substantially larger tracking, force-estimation, and interaction errors, while PIFA-NW and PIFA-NP achieve lower errors. PIFA achieves the lowest angular tracking error and end-effector interaction force, while maintaining comparable linear tracking and force-estimation errors. Under the heavier 1.5 kg payload, the baseline methods exhibit larger errors across the four metrics. PIFA achieves the lowest errors across all four metrics, reducing the angular tracking error by 18.3% compared with PIFA-NP. These results indicate that constraining the learned prediction with whole-body dynamics improves robustness to changes in payload dynamics.

Under external disturbances, DA and TD3+BC show higher tracking and interaction errors, whereas the two PIFA variants achieve lower errors but are consistently outperformed by the full model. Under torso disturbance, PIFA achieves the lowest values across all four metrics, reducing the angular tracking error by 19.1% compared with PIFA-NP. When disturbances are applied to both the torso and arm, PIFA reduces the angular tracking error by 17.3% compared with PIFA-NW. These results show that PIFA can continue to follow the human-guided motion while compensating for external disturbances, demonstrating the benefit of physics-informed prediction and forward-looking compensation for human-intent tracking and disturbance robustness.

C. Real-World Collaboration with Varying Intent and Disturbance

We validate PIFA using a physical Unitree Go1 quadruped equipped with a K1 manipulator, with onboard control running on an NVIDIA Jetson Orin computer. The experiments

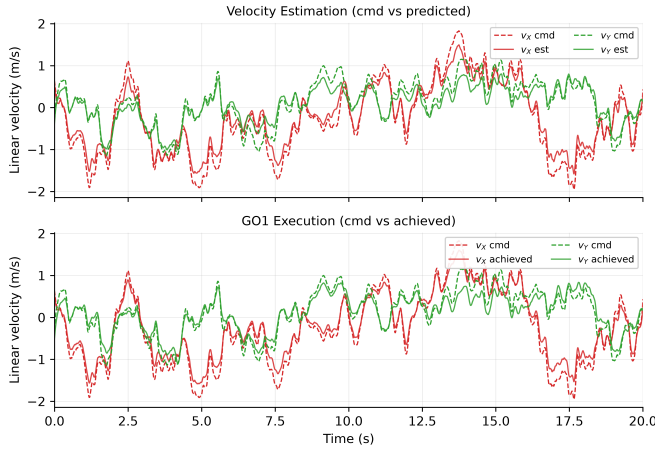


Fig. 4. Planar velocity estimation and tracking during collaborative transportation under disturbance over time. Top: commanded and estimated human-intent velocities along the x - and y -axes. Bottom: commanded human-intent velocities and achieved robot velocities along the x - and y -axes.

evaluate collaborative transportation under diverse human motion intents and terrain disturbances. From the robot, joint encoders provide $\mathbf{q}$ and $\dot{\mathbf{q}}$; the IMU provides $\boldsymbol{\theta}$, $\dot{\boldsymbol{\theta}}$, and $\dot{\mathbf{p}}$; and the onboard state estimator provides $\mathbf{p}$ and $\dot{\mathbf{p}}$ from kinematic and IMU measurements using a Kalman filter.

The qualitative results are shown in Figure 1. The top two rows show the robot’s responses to lateral, turning, forward, backward, and vertical intents. In each case, the robot smoothly follows the human-guided motion of a 0.75 kg payload while maintaining stable whole-body motion for collaborative transportation, demonstrating that PIFA can infer diverse human motion intents from proprioceptive observations and translate them into coordinated robot motion without direct force sensing.

The bottom row shows a more challenging collaborative transportation scenario, where the human guides a 2 kg payload across a bridge with slopes and steps. The robot maintains stable locomotion and continuous payload interaction despite terrain-induced body motion and external disturbances. The quadruped and manipulator jointly adapt to these perturbations, demonstrating robustness to terrain variations during physical collaboration. Overall, these results demonstrate the applicability of PIFA to real-world human-robot collaborative transportation.

D. Analysis of Intent Estimation and Motion Execution

We further analyze PIFA’s human-intent estimation and the corresponding robot motion execution. As shown in Figure 4, the top plot compares the commanded human-intent velocities with the estimated intent velocities. The estimated velocities closely follow the commanded intent across rapid changes in direction and magnitude, showing that PIFA can recover human motion intent from proprioceptive observations under external disturbances. The bottom plot compares the commanded human-intent velocities with the executed quadruped base velocities. The executed velocities closely track the commanded intent with small deviations,

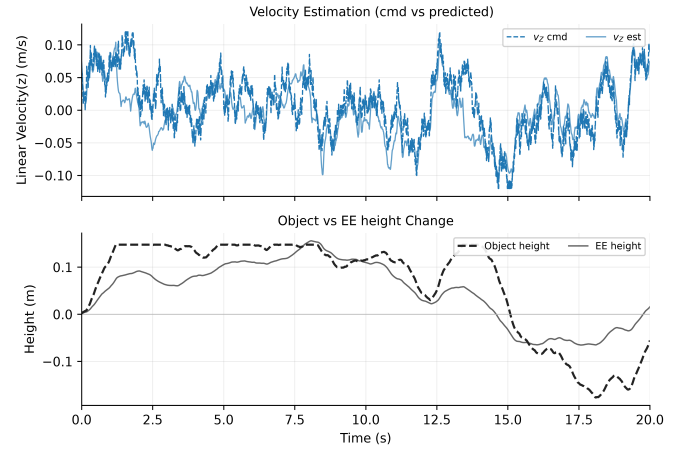


Fig. 5. Vertical velocity estimation and end-effector height tracking under arm disturbance over time. Top: commanded and estimated human-intent velocity along the z -axis. Bottom: object and end-effector heights during vertical collaborative motion.

demonstrating that the estimated intent can be effectively translated into whole-body locomotion.

Figure 5 further analyzes vertical intent estimation and end-effector motion execution under arm disturbances. The top plot shows that the estimated human-intent velocity along the z -axis closely follows the commanded vertical intent under disturbance. The bottom plot shows that the end-effector height tracks the corresponding object-height variation during vertical collaborative motion. The manipulator maintains accurate tracking performance, with only slightly higher errors due to position control and the lack of direct torque-level disturbance compensation. Overall, the quantitative analysis demonstrates that PIFA accurately estimates human motion intent and translates the estimated intent into coordinated base motion and end-effector tracking under disturbances.

V. CONCLUSION

We propose PIFA, a physics-informed forward-looking adaptation approach for intent-aware human-robot collaborative transportation. It first encodes histories of robot proprioceptive observations to capture temporal motion and interaction context and jointly predicts human intent, interaction wrench, non-intent forces, and the future robot state. These predictions are constrained by the robot’s whole-body dynamics to maintain physical consistency. It then converts the predicted intent into leg and arm commands while using the predicted non-intent forces and future robot state to compensate for disturbances. Extensive simulation and real-world experiments show that PIFA enables more accurate human-intent tracking and force estimation while reducing end-effector interaction forces under varying payloads and external disturbances compared with existing methods. One limitation is that the PIFA assumes a maintained grasp configuration during transportation. Future work will investigate adaptive grasp configurations to improve workspace flexibility, load distribution, and support more diverse collaborative transportation tasks.

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